Pi-Lambda Theorem

π-λ Theorem : If $\mathcal{C}$ is π-system, then $\lambda\langle\mathcal{C}\rangle = \sigma\langle\mathcal{C}\rangle$

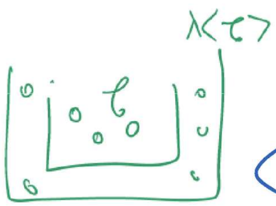
Pf: $\lambda\langle\mathcal{C}\rangle \subseteq \sigma\langle\mathcal{C}\rangle$ since $\sigma\langle\mathcal{C}\rangle$ is a σ-alg, hence a λ-system ($\because$ σ-alg) containing $\mathcal{C}$ and $\lambda\langle\mathcal{C}\rangle$ smallest such.

will show $\lambda\langle\mathcal{C}\rangle$ a σ-algebra. ($\Rightarrow \sigma\langle\mathcal{C}\rangle \subset \lambda\langle\mathcal{C}\rangle$)

$\lambda\langle\mathcal{C}\rangle$: (i) $\Omega \in \lambda\langle\mathcal{C}\rangle$ (a)

(b) (ii) $A \in \lambda\langle\mathcal{C}\rangle \Rightarrow A^c = \Omega \setminus A \in \lambda\langle\mathcal{C}\rangle$.

(d') (iii) $A_i \uparrow A_i \in \lambda\langle\mathcal{C}\rangle \Rightarrow \cup A_i \in \lambda\langle\mathcal{C}\rangle$



only need to verify:

finite unions in $\lambda\langle\mathcal{C}\rangle$. (c)

$A \cup B = (A^c \cap B^c)^c$

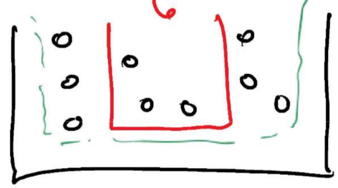
$\Longleftrightarrow$ finite intersections in $\lambda\langle\mathcal{C}\rangle$

(need to show $\lambda\langle\mathcal{C}\rangle$ is a π-system)

i.e will prove: $\forall A \in \lambda\langle\mathcal{C}\rangle,\ \forall B \in \lambda\langle\mathcal{C}\rangle \Rightarrow A \cap B \in \lambda\langle\mathcal{C}\rangle$

1st prove (1) (c) if $A \in \lambda\langle\mathcal{C}\rangle$, if $B \in \mathcal{C} \Rightarrow A \cap B \in \lambda\langle\mathcal{C}\rangle$

1st prove ① if $A \in \lambda\langle\mathcal{C}\rangle$, if $B \in \mathcal{C}$ $\Rightarrow A \cap B \in \lambda\langle\mathcal{C}\rangle$



define: (good set argument)

$$\lambda_1(\mathcal{C}) = \{A \in \lambda\langle\mathcal{C}\rangle : A \cap B \in \lambda\langle\mathcal{C}\rangle, \forall B \in \mathcal{C}\}$$

Ⓐ $\subset \lambda\langle\mathcal{C}\rangle$

Ⓒ (will show $\lambda_1(\mathcal{C}) = \lambda\langle\mathcal{C}\rangle \Rightarrow$ ① pvd)

$\mathcal{C} \subset \lambda_1(\mathcal{C})$ (since $A \in \mathcal{C}, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C} \subset \lambda\langle\mathcal{C}\rangle$)

will prove $\lambda_1(\mathcal{C})$ is a λ-system.

(why? since $\lambda\langle\mathcal{C}\rangle$ smallest λ-syst $\supset \mathcal{C}$, Ⓑ $\lambda\langle\mathcal{C}\rangle \subset \lambda_1(\mathcal{C})$)

Ⓐ + Ⓑ $\Rightarrow$ Ⓒ

(i) $\Omega \in \lambda_1(\mathcal{C})$ ($\Omega \in \lambda\langle\mathcal{C}\rangle$, $\forall B \in \mathcal{C}$, $\Omega \cap B = B \in \mathcal{C} \subset \lambda\langle\mathcal{C}\rangle$)

(ii) $A_i \uparrow$ in $\lambda_1(\mathcal{C})$ will show $\boxed{\cup A_i \in \lambda_1(\mathcal{C})}$

i.e $A_i \uparrow$, $A_i \in \lambda\langle\mathcal{C}\rangle$, $A_i \cap B \in \lambda\langle\mathcal{C}\rangle$ $\forall B \in \mathcal{C}$.

$\Rightarrow$ $\cup A_i \in \lambda\langle\mathcal{C}\rangle$ (since $\lambda\langle\mathcal{C}\rangle$ λ-syst)

$$(\bigcup_i A_i) \cap B = \bigcup_i \underbrace{(A_i \cap B)}_{\uparrow\, \in \lambda\langle\mathcal{C}\rangle} \uparrow$$

$$\boxed{\cup A_i \in \lambda\langle\mathcal{C}\rangle, \quad (\cup A_i) \cap B \in \lambda\langle\mathcal{C}\rangle}$$

$\in \lambda\langle\mathcal{C}\rangle$

$\Rightarrow \underline{\cup A_i \text{ in } \lambda_1(\mathcal{C})}$

(ii) Let $A, C \in \lambda_1(\mathcal{C})$ will check $(C \setminus A) \in \lambda_1(\mathcal{C})$

$\underline{A \subset C}$

$A \in \lambda\langle\mathcal{C}\rangle$, $A \cap B \in \lambda\langle\mathcal{C}\rangle \quad \forall \underline{B \in \mathcal{C}}$.

$+ \qquad\qquad +$

$\underbrace{C \in \lambda\langle\mathcal{C}\rangle}_{\downarrow}$, $\underbrace{C \cap B \in \lambda\langle\mathcal{C}\rangle}_{\downarrow} \quad \forall B \in \mathcal{C}$.

$\underline{(C \setminus A) \in \lambda\langle\mathcal{C}\rangle}$

$$(C \setminus A) \cap B = C \cap A^c \cap B$$

(for any $B \in \mathcal{C}$)

$$= (C \cap B) \cap (A \cap B)^c$$

$$= \underset{\in \lambda\langle\mathcal{C}\rangle}{(C \cap B)} \setminus \underset{\in \lambda\langle\mathcal{C}\rangle}{(A \cap B)} \in \lambda\langle\mathcal{C}\rangle.$$

$(A \cap B) \subset (C \cap B)$

Hence $C \setminus A \in \lambda_1(\mathcal{C})$

Hence $\lambda_1(\mathcal{C})$ is λ-syst $\Rightarrow \boxed{\lambda_1(\mathcal{C}) = \lambda\langle\mathcal{C}\rangle}$

i.e

$\underline{\forall A \in \lambda\langle\mathcal{C}\rangle, \text{ and } \forall B \in \mathcal{C} \Rightarrow A \cap B \in \lambda\langle\mathcal{C}\rangle}$

same statement as above.

* $\boxed{\forall B \in \lambda\langle\mathcal{E}\rangle, \text{ and } \forall A \in \mathcal{E} \Rightarrow A\cap B \in \lambda\langle\mathcal{E}\rangle}$

(2) ~~will prove~~ define. $\lambda_2(\mathcal{E}) = \{\underline{A \in \lambda\langle\mathcal{E}\rangle} : A\cap B \in \lambda\langle\mathcal{E}\rangle \ \forall B \in \lambda\langle\mathcal{E}\rangle\}$

$\subset \lambda\langle\mathcal{E}\rangle$

will prove : $\lambda_2(\mathcal{E}) = \lambda\langle\mathcal{E}\rangle$ (end of pf)

Note : $\mathcal{E} \subset \lambda_2(\mathcal{E})$ (from 1st step *)

($A \in \mathcal{E}$, from (*) $A \in \lambda_2(\mathcal{E})$)

Now will show : $\lambda_2(\mathcal{E})$ is a λ-sys. ($\supset \mathcal{E}$)

(since $\mathcal{E} \subset \lambda_2(\mathcal{E})$, this will imply $\lambda_2(\mathcal{E}) \supset \lambda\langle\mathcal{E}\rangle \Rightarrow$ end of pf)

(pf is same/similar as 1st part pf.)